

A fusion architecture to deliver multipurpose mobile health services

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ABSTRACT

Mobile Health (mHealth) services typically make use of customized software architectures, leading to development-dependent fragmentation. Nevertheless, irrespective of their specific purpose, most mHealth services share common functionalities, where standard pieces could be reused or adapted to expedite service deployment and even extend the follow-up of appearing conditions under the same service. To harness compatibility and reuse, this article presents a data fusion architecture proposing a common design framework for mHealth services. An exhaustive mapping of mHealth functionalities identified in the literature serves as starting point. The architecture is then conceptualized making use of the Joint Directors of Laboratories (JDL) data fusion model. The aim of the architecture is to exploit the multi-source data acquisition capabilities supported by smartphones and Internet of Things devices, and artificial intelligence-enabled feature fusion. A series of interconnected fusion layers ensure streamlined data management; each layer is composed of microservices which may be implemented or omitted depending on the specific goals of the healthcare service. Moreover, the architecture considers essential features related to authentication mechanisms, data sharing protocols, practitioner-patient communication, context-based notifications and tailored visualization interfaces. The effectiveness of the architecture is underscored by its instantiation for four real cases, encompassing risk assessment for youth with mental health issues, remote monitoring for SARS-CoV-2 patients, liquid intake control for kidney disease patients, and peritoneal dialysis treatment support. This breadth of applications exemplifies how the architecture can effectively serve as a guidance framework to accelerate the design of mHealth services.

1. Introduction

mHealth consists of medical and public health practice supported by a technological framework that leverages mobile and IoT devices to enhance communication, collaboration, data collection, and health-related activities [1]. Nowadays, IoT, combined with the broad possibilities of Artificial Intelligence (AI) techniques, can help build systems to improve medical judgement and follow-up, enhancing patient information, disease management and treatment adherence, while also supporting clinical research studies [2,3]. mHealth services usually target two primary user groups: *patients*, who seek medical evaluation, diagnosis, treatment, or advice to address their health needs; and *clinicians*, including doctors, nurses, physician assistants, and other licensed medical professionals who diagnose, treat, and manage health conditions [4]. These systems can provide tailored solutions to these user groups at different stages of health and disease, such as pre-diagnosis (e.g., prevention, health assessment, health education), diagnosis (in

particular, contributing to detect early symptoms), treatment (with the ability to analyze adherence and outcomes) or follow-up (e.g., automated risk analysis and comorbidities management) [5]. The scientific literature covers the use of mHealth systems for a wide range of conditions: breast cancer management [6], diabetes management [4], weight control [7], heart diseases monitoring [8], pregnancy care [9], overall health status [10], mental disorders [11,12] ...

It is generally assumed that mHealth may contribute to the sustainability of both public and private healthcare systems (in terms of costs [13]), to widening access to healthcare services (both in terms of location and entry into the clinical pathway [14]), and overall patient, clinician, and caregiver satisfaction [15]. By enabling continuous monitoring of patients' condition, it outspreads the reach of medical care, extending it from hospitals into the daily lives of patients [1]. Thus, in terms of effective adoption, mHealth systems are slowly becoming more prevalent in everyday life [16,17]. Many challenges remain, including privacy and security, regulation, interoperability and

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integration with EHR (Electronic Health Records), scalability and maintenance, user engagement and universal accessibility (including how to manage the digital divide) [18]. The ethical implications of using AI in these systems are increasingly emerging as a significant concern [19]. And in general, comprehensive impact analysis studies are still ongoing for many conditions [20,21] to guarantee the availability of evidence-based practices for design [22].

In commercial terms, recent statistics on the availability of healthcare and medical applications (apps) show a wide range of offerings across marketplaces (e.g., approximately 55,000 apps in Google Play in Q3 2022 [23]). From a development perspective, a thorough examination of the literature reveals that mHealth services are typically created through ad-hoc approaches, where researchers individually engineer and implement unique software architectures with distinct processing components (e.g., Refs. [10,17,18,20]). In general, the resulting designs are incompatible, leading to fragmented solutions. This high customization for specific services poses challenges in reusing existing approaches to develop new services, and further complicates their extension to meet broader healthcare needs or address new diseases within the same mHealth framework. For instance, managing the progression of a disease, which may evolve from a primary condition to one with comorbidities, may become complex if the system is not prepared to do so [24]. And in general, systems designed for the management of specific comorbidities (e.g., cancer and mental health [25,26], chronic diseases in the elderly [27], etc.) are not extensible by design. However, there appear to be significant similarities in the data acquisition, processing, and user requirements across various mHealth systems. As the reader will notice, this article intends to substantiate this hypothesis through the analysis of a sample of heterogeneous mHealth systems in the literature.

Another relevant assumption in our research is that the design of mHealth services can be approached as a fusion problem, at sensor, feature and decision-making levels. The fusion approach, in general, facilitates leveraging the strengths of different data sources (smartphone and IoT sensors, patient reported data, clinician input, etc.) to build decision support tools, while compensating for the individual limitations of the data. As will be elaborated upon, existing architectures provide a broad framework that presents challenges in translating mHealth services into actionable functionalities. These architectures often overlook the specific role of AI and neglect to consider other essential factors, such as user interaction and adherence.

Thus, the primary focus of this article is to introduce a *fusion architecture, grounded in the fusion theory*, that harnesses IoT and AI to *steer and accelerate the delivery of mHealth services*. The proposal aims to streamline design and development efforts by minimizing the time spent on commonly reusable components, considering the information flows among stakeholders, while prioritizing the core functionality or "health logic," that offers significant value to them involved parties. This contribution to the design of mHealth systems can lead to reduced development costs, more extensible solutions (in terms of functionality), easier to maintain systems and better knowledge management towards the delivery of personalized services.

The architecture hereby presented has been built following a mixed approach. First, a bottom-up inductive work has been done, including a throughout review of literature (Section 2) to provide a *mapping of functional requirements for existing mHealth services* (Section 3). Second, a domain transfer analysis has been conducted, wherein a reference fusion theory model (JDL model) has served to organize the identified requirements into a *comprehensive fusion architecture*, composed by functional microservices to leverage both IoT and AI (Section 4). And third, case study research has been used to demonstrate the *applicability of the architecture* (Section 5), wherein four real-world mHealth solutions have been designed: 1) early risk assessment for youth with mental health issues, 2) remote monitoring for SARS-CoV-2 patients, 3) liquid intake control for kidney disease patients, and 4) support to automated peritoneal dialysis treatment. The article follows this sequence,

complemented by a discussion of the similarities and differences between the proposed architecture and previous research (Section 5), and conclusions (Section 6).

2. Related work

To facilitate the delivery of an mHealth service, a collection of computing and communication systems is essential. Typically, mHealth systems leverage users' smartphone apps and IoT sensors to gather the data needed for system operation. These systems are not usually monolithic, rather they often comprise various services constituting a microservices architecture [28]. Most mHealth services in the literature use a *centralized or client-server architecture*, i.e. all the collected data are gathered on a server with the logic to generate deeper insights [29], although strict regulatory and security requirements may discourage the use of centralized architectures. As an alternative, there are proposals using decentralized architectures (e.g., based on blockchain and IPFS, a distributed storage system [4]). However, such distributed implementations are not common. The additional complexity of such systems, together with other (not always technical) challenges, discourages developers from implementing distributed approaches.

Focusing on centralized architectures, the norm is to propose highly customized architectures for specific vertical use cases or for specific challenges (e.g., securitization of data management [30] or reduced computational resources in smartphones [31]). However, despite the differences in the architectures, a set of common functionalities can be identified among all of them (as detailed in Section 3). Hence, when designing a new mHealth service, it appears inefficient to overlook the previous knowledge regarding commonalities among existing services and opt for a design starting entirely from scratch. In this direction, some authors before us have already addressed the development of generic mHealth architectures that can be adapted to different use cases. For example, the client-server architecture in Ref. [32] consists of 4 layers: user layer (i.e., user interaction), communication layer, application layer (i.e., health functionality) and middle layer (i.e., data storage, authentication). A layered approach is also proposed in Ref. [9], with a core layer managing data and relationships, an application layer managing APIs and user interaction, and an environment layer considering integration with external systems. The proposal in Ref. [33] shows a flexible architecture, where part of the data pre-processing is delegated to mobile nodes. It also identifies generic nodes for communication, data aggregation, data processing and visualization. Authors in Ref. [34] recognize the value of cloud platforms for implementing mHealth systems. Their proposal includes a layer to manage multi-tenant cloud storage and access control, combined with data annotation and data analysis layers. In Ref. [35], a framework for developing a mHealth system is presented, based on the implementation of several functionalities: data acquisition, health knowledge extraction, persistent data storage, adaptive visualization, system management and other optional services, such as intelligent alerts, recommendations and guidelines. This architecture focuses on data collection and monitoring functions. The most comprehensive flexible architecture found in the literature is the one presented in Ref. [36] and developed by IBM. It consists of an initial data ingestion layer, a storage layer, a data analysis layer, and a data access layer, all supported by common security and access control services. It also considers the existence of several user interfaces for data interaction and visualization.

Thus, most of the existing flexible proposals advocate structuring the architecture into a set of layers. Likewise, it also stands out that along with the need for a set of layers dedicated to data capture and analysis, proposals detect the need of layers dedicated to interaction, storage or user management. While these architectures may serve as initial guides for designers and developers, they remain being relatively high-level ones, lacking detailed specifications required to implement the targeted mHealth system. *A complete identification of the blocks in each layer would help to develop common components* that could be easily shared and

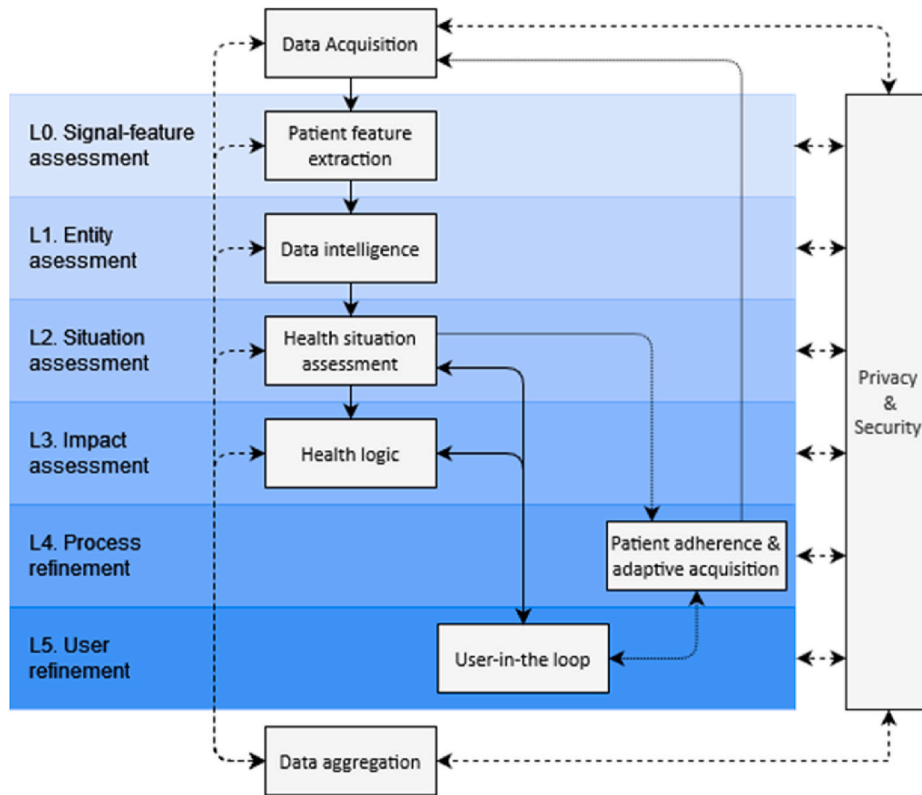


Fig. 1. Correspondence between the JDL fusion model and the proposed fusion model for mHealth service analysis.

reused.

Our proposal also adopts the layered approach, yet it surpasses the current state-of-the-art by redefining it and elaborating modular services within each layer. These services facilitate the design and development of heterogeneous services supporting various *Health Logics*. Following the initial review of architecture approaches in the current section, the next section compiles a bottom-up review of systems to extract the most common mHealth functionalities into a comprehensive mapping.

3. Mapping common functionalities of mHealth systems through a data fusion perspective

At their core, *mHealth systems can be considered as a data fusion process aimed to deliver health-related functionality* [37]. First, relevant data is gathered from smartphones and IoT devices. Multiple sources of information are integrated within a single platform into a common health records storage system. Then, this aggregated information is fused in different stages into valuable health metrics using data analysis techniques. Results are evaluated to support automated or semi-automated decision making. Finally, the information is provided to end-users through different visualization strategies. Therefore, in this section, the suitability of this fusion approach is further elaborated (subsection 3.1) as a basis for extracting and explaining common functionalities among services (subsections 3.2 and 3.3) that will support our subsequent proposal of fusion architecture.

3.1. Data fusion for mHealth service analysis

Following on this idea, the subsequent objective is to identify the needs of a mHealth system by using a customized data-fusion model based on the JDL one [38]. In this model, fusion is performed gradually across 5 different levels, which increasingly broaden the scope and actionability of the information produced. *Level 0 (signal-feature assessment)* preprocesses the source information to align data, standardize

inputs and generate valuable features. *Level 1 (object assessment)* links the previous features to estimate an individual entity attributes and state. *Level 2 (situation assessment)* aggregates information from several entities to estimate the relationships between them and to recognize global situations and structures. *Level 3 (impact assessment)* projects the situation information into the future to provide consequence analysis by assessing the impact of different courses of action (it constitutes a valuable decision support tool). *Level 4 (process refinement)* can be understood as a metaprocess that continuously evaluates and manages the fusion chain (i.e., adaptive data gathering) to improve systems performance. Finally, *Level 5 (user refinement)* connects the user to the fusion pipeline, enabling fused information visualization and feedback and control of the process.

Although JDL model was originally envisioned for decision-making in military environments, it has also been applied to other domains (remote sensing, cybersecurity, financial analysis, environmental monitoring, etc.). It can also be applied in the health domain, as it resembles the clinical method used by clinicians. In this sense, the clinical process starts by gathering information from the patient about his condition (e.g., symptoms, vital signs, life patterns). The clinician intrinsically translates patient input into standardized medical attributes and filters out non-relevant information (level 0 fusion). Then, health attributes are analyzed jointly to identify anomalous attributes and to provide a diagnosis on each of the patient individual conditions (level 1 fusion). Patient comorbidities are considered globally together with the health records to obtain a holistic view of the patient health (level 2 fusion). Based on this view, clinicians can provide a prognosis and to identify different treatments or courses of actions (level 3 fusion). Throughout this process, additional information may be required to refine the diagnosis in the form of complementary medical tests and to monitor patient evolution (level 4 fusion). Additionally, clinician-patient communication is critical (level 5 fusion). Thus, the hierarchical nature JDL fusion model (it builds a holistic view from individual attributes) together with its comprehensiveness (it covers all the steps in

Table 1
Analysis of existing literature and identification of common features across mHealth systems.

Ref.	Feature group	General features						mHealth fusion features			
		Target users			Supported devices		Security	Data acquisition		Data aggregation	
		U1	U2	U3	D1	D2	NF1	AC1	AC2	AC3	AG1
		Patient	Clinicians	Family	Smartphone	Web access	Security & Privacy	Self-reported data	Embedded sensors	IoT wireless devices	Data integration
[4]	Diabetes management	✓	✓				✓		✓	✓	✓
[77]	Diabetes self-management	✓			✓		✓	✓	✓	✓	✓
[50]	Diabetic retinopathy diagnosis	✓	✓		✓			✓		✓	✓
[55]	Chronic kidney disease self-management	✓			✓	✓		✓		✓	✓
[42]	COVID-19 detection	✓	✓					✓		✓	✓
[65]	COVID-19 management	✓	✓					✓		✓	✓
[78]	Epidemic control system	✓	✓		✓	✓		✓			✓
[53]	Heart disease prediction		✓							✓	
[33]	Heart monitoring	✓	✓		✓	✓	✓		✓	✓	✓
[45]	Arterial fibrillation monitoring	✓			✓		✓			✓	
[68]	Hypertension management	✓	✓					✓			
[69]	Hypertension monitoring	✓	✓		✓			✓			
[8]	Hypertension management	✓	✓		✓	✓	✓	✓	✓		
[7]	Obesity and hypertension management	✓	✓		✓			✓		✓	✓
[62]	Pregnancy risk detection	✓	✓	✓						✓	
[9]	Pregnancy care management	✓	✓		✓	✓		✓		✓	✓
[44]	Pregnancy monitoring	✓	✓		✓	✓	✓	✓	✓	✓	
[6]	Breast cancer	✓	✓		✓			✓			
[79]	Pediatric obesity treatment		✓	✓	✓	✓		✓		✓	✓
[10]	Overall health self-management	✓						✓			
[35]	Overall health self-management	✓			✓			✓	✓	✓	✓
[61]	Physical and emotional health support	✓			✓			✓	✓	✓	✓
[52]	Physical activities of elderly people		✓	✓			✓			✓	✓
[64]	Hip fracture telerehabilitation	✓	✓	✓	✓	✓					
[36]	Alzheimer monitoring	✓	✓		✓			✓	✓	✓	✓
[49]	Food intake management	✓			✓			✓			
[47]	Food intake management	✓			✓			✓			
[48]	Food recognition for visual impaired children	✓			✓			✓			
[73]	Dietary management	✓	✓		✓		✓	✓		✓	
[46]	Caries-related diet tracking	✓	✓		✓	✓		✓			✓
[70]	Dental diagnosis	✓	✓		✓	✓	✓	✓			
[41]	Suicide prevention	✓	✓		✓		✓	✓	✓	✓	✓
[11]	Bipolar disorder assessment	✓			✓			✓	✓		✓
[12]	Post-traumatic stress disorder monitoring	✓	✓		✓		✓	✓			✓
[41]	Suicide prevention	✓	✓		✓		✓	✓	✓	✓	✓
[72]	Smoking Cessation	✓			✓			✓			
Aggregated occurrence (over 36)		33	25	4	28	10	12	29	11	21	20

the clinical process) makes it suitable for the healthcare domain.

As mHealth services assist both patients and clinicians through the clinical process, they implement a data fusion chain covering the previously defined levels. That is, they enable health data acquisition, relevant health features extraction, intelligent data analysis and diagnosis, comprehensive patient health assessment, implementation of different health logic (ranging from prognosis estimation, recommendation generation, health events notification), and health data visualization. Thus, JDL model can constitute a valuable conceptual framework to analyze the features and requirements of mHealth services. In this sense, we propose an adapted and extended version of the JDL framework (as shown in Fig. 1). This model includes additional levels for data acquisition and data aggregation, which are critical in mHealth services, while also considering privacy and security issues. To our knowledge, there is no similar conceptual framework that enables analyzing mHealth services while considering a holistic evaluation of the patient health.

3.2. Common features by fusion layer

Considering this health-centric fusion model, a systematic review of more than 50 mHealth services in literature has been conducted, layer by layer, to identify common features. The study covered systems spanning from 2011 to 2024, enabling a comprehensive analysis of various scenarios and constraints. This breadth aims at obtaining an overview of features that are generalizable. Articles included in this review satisfy the following criteria: 1) define a system that can be used to improve a healthcare situation; 2) further elaborate on the data collected; 3) develop the functionalities of the system and its objectives; 4) indicate the outputs of the system and the interaction of users with it. Of all these articles, those better fulfilling the imposed criteria have been included in Table 1 and further evaluated. These have resulted in a deep analysis of 36 mHealth systems considering 26 conditions.

Throughout the ensuing discussion, the identified *features* are denoted by *identifiers in square brackets*, and examples from the literature

mHealth fusion features										
Data aggregation	Feature extract.	Data intelligence			Health sit. assessment	Health logic	Adhe-rence	User-in-the-loop		
AG2	FE1	DI1	DI2	DI3	HA1	HL1	AD1	UL1	UL2	UL3
EHR Interoperability	Data preprocessing	Anomaly detection	Inference engines	AI solutions	Health State evaluation	Rule-based functionality	Adherence support	Notifications	Communication	Data visualization
✓	✓									✓
	✓									
	✓						✓	✓	✓	
	✓	✓		✓	✓		✓	✓		✓
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are cited accordingly. For those interested, a detailed description of each layer and its respective features, is provided next, supported by literature examples. Additionally, readers seeking a summarized analysis can refer directly to Section 3.3.

3.2.1. Data acquisition

A precondition for data fusion is to gather relevant data from heterogeneous devices, which can be useful for patient’s diagnosis, treatment, or follow-up. Smartphones are usually in charge of data acquisition. They are used to the *active sending of patient data* [AC1], including information directly input by users through forms or questionnaires (e.g., their self-perceived health status or mood [11]), or measurements from non-connected devices (e.g., glucose levels [39]). Additionally, *sensors embedded* [AC2] in smartphones can provide continuous and non-invasive contextual data such as location, social interactions (e.g., messaging, social networks usage or phone communications statistics [40]) or behavioral patterns (e.g., physical activity or

sleep [4]). Finally, smartphones can also work as a hub to external IoT/wireless devices [AC3] (e.g., wearable ones, such as activity bands [41], or non-wearable ones, such as scales or infusion pumps [42]). Raw sensor (embedded or wireless) data does not provide semantics and therefore cannot be automatically interpreted by a computer system [4]. In this sense, the Semantic Sensor Network (SNN) ontology describes relevant concepts and relationships for sensors and their data. It allows for the representation of IoT data records and facilitates integration with other ontologies and semantic data models [43]. IoT sensors can measure a single biometric variable (homogeneous), two variables (dual) or multiple variables (heterogeneous), the latter being the most common and effective [37]. As well, mHealth systems can receive data from other sources besides smartphones, such as clinician reporting (via web interfaces) or hospital medical records. Moreover, Machine Learning (ML) and Deep Learning (DL) algorithms can be employed to analyze the quality of raw data and estimate missing data [44].

3.2.2. Data aggregation

Once data is acquired, *data aggregation and integration* [AG1] is a requirement to successfully perform the data fusion process. This includes supporting elements such as methods of data management, multi-source data merge and data storage, and strategies to integrate data into internal hospital systems. A centralized platform (e.g., cloud computing platform) is usually required to interconnect the final devices and to provide both storage and analysis resources. Even in cases where a device-centered approach is used (e.g., health self-management with limited onboard analysis [10]), a centralized computing platform is still used for other relevant tasks (e.g., send personalized notifications [39]). To achieve efficient data integration within databases, standards for communication protocols and data formats and management systems, allowing the collection, integration and analysis of data from different sources are beneficial [9]. It is expected that both structured and non-structured data stores might be used depending on the type of information to be recorded. Stores might be deployed on premise or on cloud service providers, as is the case of [45].

For the moment, the use of mHealth systems remains silo oriented, so they are often not fully integrated with existing Hospital Information Systems (HIS), which hold the registry of medical records, patient data and organizational information. However, full bidirectional *interoperability with Electronic Health Records* (EHR) [AG2] would enable the integration of mHealth-retrieved data with traditional hospital processes. Multiple standards can be used, such as the HL7 FHIR (Fast Healthcare Interoperability R43resources) which enables the exchange of information between a wide range of different healthcare systems and health monitoring devices [43]. It also supports the use of web technologies (e.g., REST and JSON) facilitating integration with diverse systems [36].

3.2.3. Patient feature extraction

Collected data can be initially used to extract relevant features. Literature shows that a preliminary *data preprocessing* [FE1] can remove environment noise and motion artifacts that may have been introduced during data acquisition [35,44]. Then, data can also be processed to generate interpretable features (entities, patterns or events with a given meaning or possible medical relevance) which can be later used to make further analysis or take health-related decisions. For example, the image processing can be used to detect food in a picture (suitable in case of dietary requirements [46,47] or for visually impaired patients [48]), to extract food's nutritional features [49] or to extract features that help with diseases diagnosis (e.g., for diabetic retinopathy detection [50]). Also, sensor data can be processed to obtain interpretable features (e.g., average heart rate [4]).

Processing often takes *advantage of AI solutions* [DI3]. For example, the use Convolutional Neural Networks (CNN) for image processing is widespread, either self-developed networks [48,50] or using existing networks (e.g., Xception [46], EfficientNet [47], YOLO [49], VGG [51]); transfer learning is applied to improve accuracy in the explicit domain. DL can also extract complex information from large volumes of raw data such as quality sleep features from vital signals [41] or movement patterns from IoT data [52].

3.2.4. Data intelligence

The previously collected features can be used to obtain information related to the patient's health. One way to obtain this is through the *anomaly detection* [DI1], which can refer to detecting anomalous values or relevant changes in the evolution of features. This may include comparison with thresholds that indicate risk if exceeded (e.g., heart rate value to detect tachycardia or bradycardia [53], or blood pressure to detect hypertension and hypotension [54]), or further data analysis to identify anomalous patterns [34] (e.g., detect kidney issues based on illness patterns evaluation [55], detect variations in medication adherence or sleep quality based on IoT data [33], or eczema detection by classifying features extracted from images [51]). Also, an *inference*

engine [DI2] can be used to obtain more sophisticated health features, such as breathing state identification based on seismocardiogram (SCG) and electrocardiogram (ECG) signals [56].

Such approaches can be assisted by the application of ML, data mining or *AI solutions* [DI3] reducing significantly computing costs and time [57] and obtaining more accurate results [58]. AI can be used to learn from the data collected, automating analysis processes (by combining statistics and classification) and generating customized information that can feed decision support systems [19]. Regarding anomaly detection, AI solutions allow to identify changes or negative patterns against the patient historical data [58]. Some ML algorithms useful for anomaly detection are Bayesian Belief Network (BBN), K-Nearest Neighbors (KNN), Self Organizing Map, SVM, Naïve Bayes, Random Forest or AdaBoost, which can also be combined with gaussian statistical models to improve their precision and efficiency [51,59]. Anomalies can also be detected on images using AI, such as CNN for skin cancer detection [60]. In terms of obtaining patient features, algorithms such as Decision Trees are used to predict emotions based on user's voice records [61] or phone sensor data (sleep patterns physical activity and frequent locations [40]), Naïve Bayes to identify exercises performed by patients combining IoT data [52], CNN such as multilayer perceptron to detect bipolar disorders [11] or hybrid systems, combining ML/DL with knowledge, to determine levels of stress in pregnancy from knowledge of how stress affects health metrics [44].

3.2.5. Patient situation assessment

Deepening on the fusion chain, mHealth systems may perform a *health state evaluation* [HA1] by obtaining a comprehensive health status picture, evaluating patient's risk to a given disease [57] or developing predictive models for early detection of diseases and changes in health [37]. Relevant information (e.g., key health indicators, health risk) that may be useful for posterior decision making (e.g., recommendations, diagnosis) is extracted.

Literature examples include the use of previously detected anomalies to estimate the risk of suffering heart disease [33] or kidney disease [55]. ML/AI can also be used to develop early warning algorithms for a wide variety of diseases, such as risk detection in pregnancy using Support Vector Machine (SVM) [62], Logistic Regression to predict relapse in mental health based on mobility patterns changes [41] and Natural Language Processing (NLP) can be performed on audio to detect diseases (e.g., Alzheimer [36], mental health conditions [61], asthma [63] or COVID-19 by cough worsening [42]). Although the use cases reviewed are generally focused on specific diseases, this layer could also perform an integrated comorbidity analysis to create an overall picture of the patient's health.

3.2.6. Health logic

Patient assessment can ultimately be used to support the decision making and to implement the health logic of each system. It can encompass patient recommendations (e.g., lifestyle suggestions and alerts) and consider the generation of alternative clinical or diagnosis courses of action (CoA, in fusion terminology), by analyzing trends or evolutions, feeding the patient history with curated data, so decisions are consistently supported over the patient evolution [64]. It can also enable different workflows depending on the health status, so proportional actions with human-in-the-loop (e.g., alerts) can be carried out [65]. Overall, health logic can generate varied actions to fulfil different system goals (e.g., improving treatment or drugs adherence [66], educating patients [67], enhancing clinical care [68], advance health risks [68], or conducting clinical research [36]).

To handle part of these functions, it is usually needed to count on *rule-based functionalities* [HL1], which executes a set of predefined rules based on specific conditions. Logical conditions and actions can be implemented to decide when to deliver a notification, to classify the severity of an alert [65] or to decide which information is displayed [69]. Expert systems, case-based reasoning and other AI tools may also

be used to implement the core health logic of the system. Algorithms for decision support can be used to take the above-mentioned actions based on patient information and provide real-time feedback [61].

3.2.7. User-in-the-loop

User-system interactions can empower clinicians as decision-makers in semi-automated processes integrated into the clinical pathway [70]. This can be achieved through *communication* [UL2] between clinicians and patients or caregivers, through calls, chats [9] or appointment scheduling [71]. Patients can also receive *notifications* [UL1] related to reminders, alerts or communication with clinicians. *Data visualization* [UL3] (displayed as graphs [4], health indicators [55], useful information such as relaxing exercises [72] ...) is also key in mHealth systems. To ensure effective information delivery, it is important to select meaningful data, while considering vocabulary adaptation and ease of interpretation [69]. To ensure high usage of the system, it is important to provide a sound and solid user experience, considering accessibility or device portability. AI can also be used to better interact with the system users, enhancing adherence and user experience. For example, including a chatbot to provide companionship to mental health patients [11], personalized motivational messages delivery according to patient features [72], or automatic speech recognition to simplify the recording data process [73].

3.2.8. Patient adherence and adaptative acquisition

Accurate and continuous data collection is crucial to make the system knowledge base grow. Therefore, literature considers the usage of management processes to ensure sufficient data acquisition for the system's effectiveness. This can involve passive sensor, active user-generated or device-captured data [41,70]. This requires strong user adherence to the system: the degree to which users or individuals follow and engage with the intended use, guidelines and recommendations of the digital system [66]. *Adherence* [AD1] encompasses factors like usage frequency, consistency, guidelines follow-up, task completion, engagement, and data collection. Patient engagement improves system efficiency, enhances patient-doctor communication and improves treatment/medication adherence and overall patient health [69]. Examples within this layer include adapting the notifications delivered to the patient or the visualization of data (e.g., including gamification models to enhance user interaction [69]) to improve adherence. Meanwhile, adaptive acquisition refers to the intelligent adjustment of non-active data collection.

3.2.9. Non-fusion additional feature: privacy and security

Nowadays, security and privacy [S1] features are implemented by-design. Security refers to the protection of information systems from unauthorized access, disclosure, alteration, destruction, or disruption, while privacy refers to the right of individuals to control the collection, use, and disclosure of their personal information. Although they are different concepts, they are usually addressed through combined technology approaches. For example, in the EU, the General Data Protection Regulation (GDPR) is primarily concerned with privacy rather than security, although it does address security as part of its broader framework for protecting personal data.

Securing patient data may rely on encryption techniques [32], e.g. MD5 message-digest algorithm can be used to protect chat data for external attacks [73], using public and private key encryption to protect user data during transmission and storage. Blockchain techniques can also be used for this purpose, as well as providing transparency and traceability in data management. Additionally, to ensure privacy and security, personal data may be stored independently of medical data [4, 8].

Also, the regulations of each region may require an informed consent from the patient to exchange health information [74]. Therefore, in some cases, clinicians will only be able to access patient data with the patient's authorization during the clinical visit [73]. Thus, it is necessary

to provide authentication techniques so that only authorized healthcare professionals have access to patient information. Another solution to this barrier is the use of Health Record indexes, which may allow the practitioner to obtain the necessary information in an emergency, even if the patient is incapable of giving consent: the proxy informs practitioners about the health record location in the cloud, without directly providing the data [74].

In addition to these measures, an approach such as federated learning can be implemented to further improve data privacy and security [75]: under this paradigm, data is spread among different (even mobile) servers and never exchanged, as only the results of data processing layers are shared with other services. Literature also suggest including solutions against attacks, such as the use of intrusion detection techniques, data poisoning and protection against malware [76]. Similarly, backup and recovery systems to ensure the availability of data in case of a failure or catastrophic event can also be considered, usually through storage usage [34,76].

3.3. Result: A functional map for mHealth services

Table 1 serves to summarize the identified services, highlighting fusion model features and other general aspects (as discussed in Section 2, such as target users or supported devices) found within each case. Finally, the table also includes, in its final row, the relative frequency of each feature across the 36 articles included in the table. This indicator underscores the prevalence of these features, supporting their inclusion in our final architecture.

4. A multipurpose fusion mHealth architecture

Considering the common features identified in Section 3.2 (Table 1) and described in Section 3.3, an architecture is proposed, integrating the previously established health-oriented data fusion model and harnessing the capabilities of AI and IoT within its workflow. The architecture (depicted in Fig. 2) is service-oriented: relevant services have been identified encapsulating related functionalities in each layer. This approach enables modularity as only those services that are relevant for each mHealth implementation are required. Overall, the architecture aims to be a concept tool to guide the development of mHealth systems. It intends to help developers to identify and implement common reusable components that fulfills usual mHealth systems features. As a result, it enables developers to focus on the "health logic" and in the AI analysis algorithms.

Next, the architecture-level data workflow will be discussed (Fig. 3) identifying the services in each layer (Table 2 and Table 3) and highlighting their features. A brief description of the services and associated development technologies is also provided. It is important to note that the architecture does not impose any specific implementation or protocol to use, this is finally decided in the light of the mHealth context.

4.1. Architecture components and workflow description

First, data acquisition is integrated into the architecture, which is divided into two layers: the *Data Acquisition Layer* and *Data Harvesting Layer*. The first serves as a hub to collect sensors data or actively reported data. This layer includes apps and desktop interfaces that allow requesting and retrieving data directly from users. Moreover, data gathered from smartphone sensors is collected through the mobile health app, while data from IoT devices can be collected through the app itself (interaction 1a of Fig. 3) or from external servers, through APIs provided by IoT vendors (interaction 1b of Fig. 3). Meanwhile, the second centralizes all the collected data. The data received by the smartphone is handled by the internal collection service. This layer also considers the mentioned existence of an external collection service that connect to these platforms via SDKs provided by manufacturers to retrieve data. Once all the raw data is received, it is passed through an

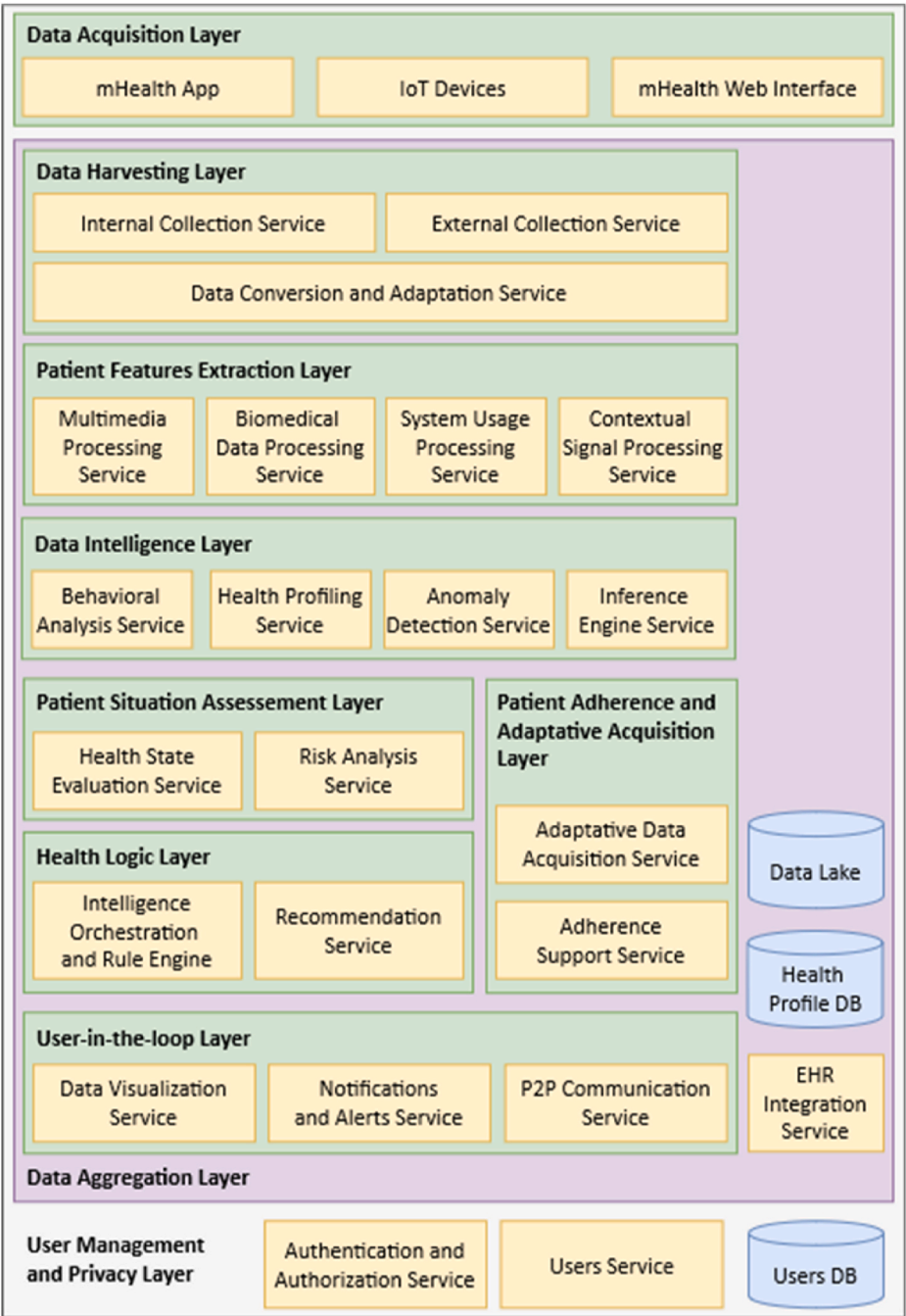


Fig. 2. A comprehensive architecture for mHealth services design based on the adaptation of JDL model.

adaptation and conversion service to maintain a uniform format and ensure integrity. Then, this data is stored in a data lake, as can be observed in the interaction 1c of Fig. 3. The data lake can be a distributed one or a service composed of various storage components deployed in different locations to decentralize information, thereby enhancing data privacy and security and facilitating the accomplishment of regulations.

To enable the fusion and exploitation workflow, the *Data Aggregation Layer* includes the common features that allow the storage and integration of the different data sources. In this sense, the layer includes storage resources for raw structured data representing the patient's health status (health profile DB), allowing access whenever needed for presentation or analysis. In both cases, information can be stored using ontologies to provide an integrated view of the patient status. Both

resources are used transversally across the layers to access and aggregate information. These databases (DBs) are decoupled from the identity of the users for patient privacy and security reasons. This solution also enables the data anonymization, providing or not the user information with the health data to the rest of the system (or to a subset of its services). Finally, this layer also allows the integration of the platform with the HIS through the EHR integration service. In this way, the health information generated by the system can be consulted from traditional systems, facilitating the work of clinicians. Similarly, hospital diagnostic tests or the patient's electronic medical history can be incorporated into the platform, enriching the analysis performed. These three layers are summarized in the following table, showing useful technologies for their implementation.

Starting with the data analysis, the *Patient Features Extraction Layer*

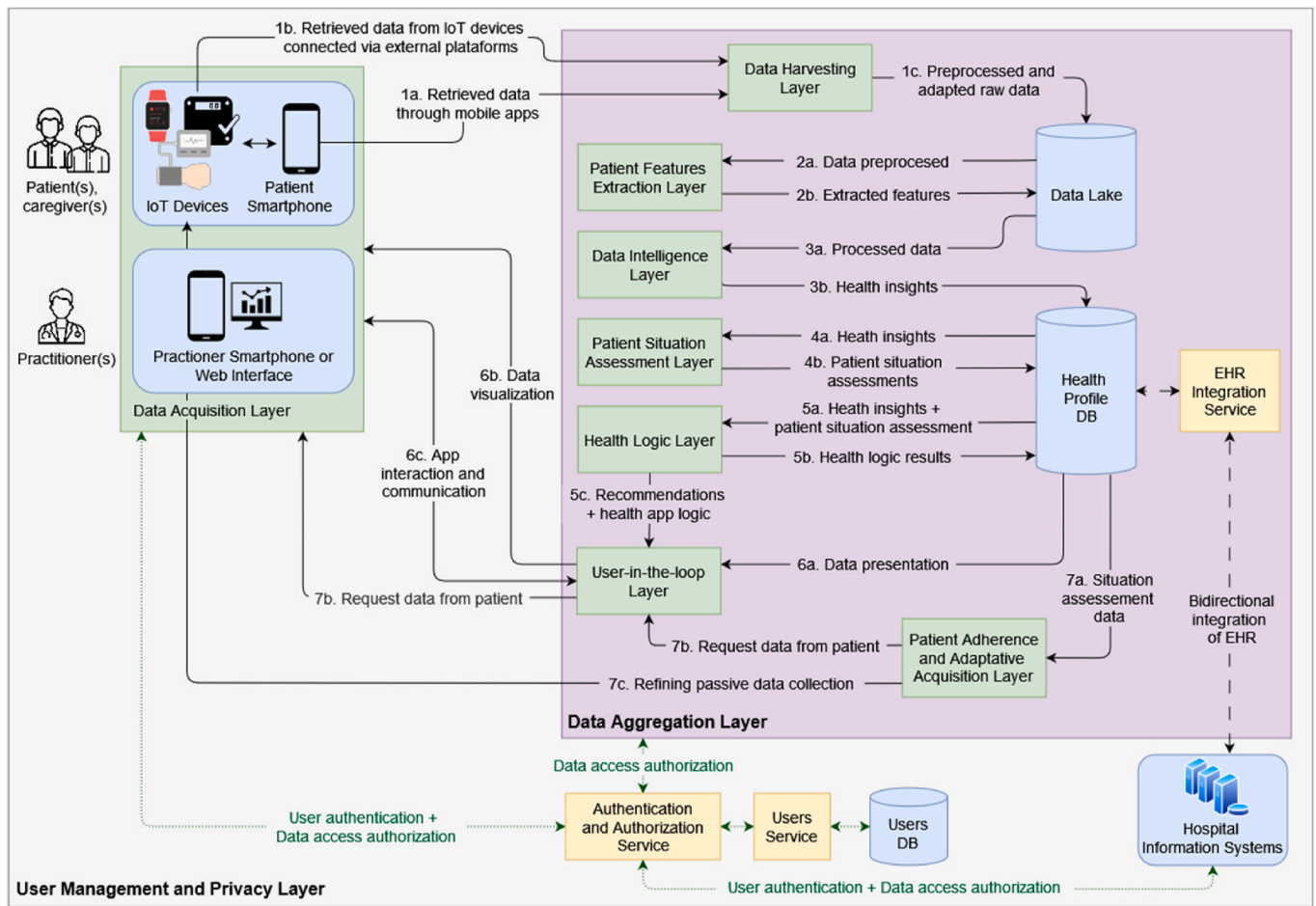


Fig. 3. Functional workflow across the architecture's layers.

enables domain-specific processing thanks to independent services oriented to different types of data (as each type requires specific techniques appropriate for it): multimedia, biomedical, system usage and contextual. All these services can allow the implementation of AI functionalities, such as object detection and classification, generation of aggregated statistics or recognition of events in biomedical data (e.g., cardiac events or glucose level alterations) or geocoding of positions. Federated learning approaches can be used to reduce the access to raw data by different services, increasing the security and preserving the privacy. To carry out these analyses, the different services obtain the previously preprocessed data from the Data Lake (interaction 2a in Fig. 3) and store the extracted features in the same database (interaction 2b in Fig. 3).

Next, the *Data Intelligence Layer* encompasses different services focusing on advanced analysis: complex information estimation (inference engine), anomalies detection, patient behavior analysis or health profiling. These services perform specific AI-aided techniques. In general, this layer generates valuable metrics that will enable evaluating the overall patients' health in the next layer. This analysis is fed by the information already processed and stored in the Data Lake (interaction 3a in Fig. 3) and it stores the obtained health insights in the Health Profile Database (interaction 3b in Fig. 3).

Following through the proposed data fusion chain, the *Patient Situation Assessment Layer* is articulated by services which consider health status metrics, real-time detected anomalies, historical data and medical records (interaction 4a in Fig. 3) to provide a comprehensive view of the patient's health and to estimate risks (interaction 4b in Fig. 3). The obtained health profile is persisted within the aggregation layer so that it can be visualized by users for decision-support.

Finally, the *Health Logic Layer* constitutes the higher level of the fusion chain, including decision support elements based on information from previous layers (interaction 5a in Fig. 3). The intelligence orchestration and rules-based engine service enable the integration of all AI-assisted analysis services of the previous layers to execute the health logic of the system. In this way, it allows configuration and coordination of analyses of different layers (e.g., combining the inference engine service and the risk analysis service to detect patients with a high probability of suffering complications). At the same time, this service can create actions according to the needs, such as high-risk alerts, periodic events, interface adaptations, etc. This information is also stored in Health Profile Database (interaction 5b in Fig. 3). Recommendations and CoA generation are also provided (interaction 5c in Fig. 3) with a dedicated service based on the acquired information.

The *User-In-The-Loop Layer* considers services oriented to provide data visualization (interaction 6b in Fig. 3, which offers personalized data presentation, dynamic dashboards, advanced filtering, etc.), P2P communication, and notifications or alerts (interaction 6c in Fig. 3, which allow remind the patient to take medication, report a problem, notify new messages, etc.). The behavior of these services is closely linked to the implemented health logic.

The management and adaptation of the data acquisition layer is supported thanks to the *Patient Adherence and Adaptive Acquisition Layer*. This layer assesses the existing data needs (based on the situation assessment received from interaction 7a of Fig. 3) and modifies the collected data, potentially activating new sensors, increasing collection frequency (interaction 7c of Fig. 3), or requesting additional information from patients or clinicians (interaction 7b of Fig. 3, which transits through the User-in-the-Loop layer to deliver it to the interfaces). It also

Table 2

Services of layers responsible for the data gathering and lower-level data management. Notes on implementation include some potential technologies/protocols that can be used when instantiating the architecture for a specific use case.

Layer	Service	General/ Fusion Features	Description	Notes on implementation
Data Acquisition	mHealth App	U1, D1, AC1-AC3	iOS or Android native app (e.g., final data visualization, communication, data retrieval) and serves as hub to gather info through the mHealth system.	OS-dependent APIs to acquire data from accelerometers, light sensors or positioning engines. App usage statistic or communication statistics may also be retrieved using virtual probes.
	IoT devices	D1, AC3	Wireless health devices that provide health metrics, such as smartwatches, smart-scales, oximeters, sensor clothing, sphygmomanometers ...	Manufacturer specific APIs and SDKs can be used to retrieve data on NFC, Bluetooth or Wi-Fi. Standard protocols such as Matter may be used for health devices.
	mHealth web interface	U1, U2, D2, AC1	Clinician interface for information retrieval describing patient's self-perceived health and treatment details, data visualization and graphical insights generation, communication, treatment update, etc.	Web development frameworks such as React or Angular can be used to develop this interface.
Data Harvesting	Internal gathering service	AC1-AC3	Centralizes data obtained from the architecture internal interfaces (i.e., mHealth app and mHealth web).	On REST or streaming-based technologies (e.g., Kafka). To ensures data integrity, it may include outlier deletion, duplicate data deletion, time alignment, etc.
	External gathering service	AC3	Retrieval from external IoT, vendor-specific platforms.	Manufacturer specific APIs and SDKs when direct connection to device is not feasible.
	Data conversion & adaptation	AG1	Data standardization.	Ontologies (e.g., SNN), may be used to associate sensor data with interpretable concepts.
Data Aggregation	Data lake	AG1	Database/s, centralized repository for storing large amounts of multimodal raw data (patient's data, medical records, diagnostic images ...).	Amazon S3, Azure Data Lake Storage and Google Cloud Storage are examples of data lakes provided as cloud-service.
	Health profile DB	AG1	Database/s containing user-linked electronic health profile and evolution: risk level, adherence to treatment, behavioral analysis, adherence to system, etc.	InfluxDB is a time series database, while MongoDB or PostgreSQL are example of OODBMS that can store health profiles efficiently. Cloud-service available.
	EHR integration service	AG2	Provides a bidirectional communication with the hospital services and databases, adapting format for compatibility and usability.	Standards (e.g., LONIC, HL7 CDA or HL7 FHIR) can be used to simplify clinical information transcription. Also, there are standards specialized in different areas of medicine, such as DICOM for medical images (e.g., radiology, cardiology imaging, and radiotherapy) or SNOMED CT for diabetes insipidus terminology.

includes an adherence support service to encourage ongoing app use, reminding patients to actively provide information or to follow the recommendations and treatments.

As a supporting, transversal layer, the User Management and Privacy Layer ensures that patients' information can only be viewed by themselves or their practitioners. To do so, the users service manages existing users and roles including sign up to the platform and log in the different applications. The authorization and authentication service assigns users an authentication key associated with their specific permissions, which is used across the platform to guarantee that a user only access authorized information. Similarly, services within the platform are also registered in the service and use keys to access and decrypt information. Optionally, further enhancing privacy, the service also allows generating one-use time-limited access codes to ensure that patient's information is accessible only by assigned professionals, under their authorization.

As explained above, AI can be applied across layers in the vast majority of services, from low-level feature extraction to high-level support to recommendations or courses of actions, going through risk analysis, behavioral analysis or profiling for different purposes. The tools mentioned include predictive algorithms, anomaly detection algorithms, clustering algorithms, knowledge integration systems for decision support, automation of multimedia content recognition or generative systems. The objectives of the specific use case determine which of these technologies need to be implemented in which service.

4.2. A comparative analysis

Table 4 shows a comparative analysis of the proposed architecture with existing architectures in the literature. Both differences and similarities between each architecture pair (proposed architecture vs literature architecture) are identified.

Even if most architectures cover the basic services for data acquisition, processing, and visualization, some relevant features of our proposal are missing. First, analyzed architectures do not identify functional components related with mHealth needs. In this sense, instead of identifying specific services (e.g., biomedical data preprocessing, risk analysis, holistic health analysis ...) dedicated to tasks related to the mHealth functionality to be achieved, generic processing services are proposed. This lack of specificity makes it difficult for the developer to instantiate the architecture and identify reusable components.

Similarly, AI-assisted processing is only considered in an architecture without identifying possible algorithms, which makes difficult to take advantage of synergies in the development of these algorithms. Another barrier is the lack of adaptive data acquisition or adherence modules which are vital to ensure that acquired data is sufficient and fulfills the requirements to achieve the mHealth logic. Finally, inter-user communication is only considered in the last analyzed architecture. Overall, it can be concluded that the proposed architecture goes beyond the state-of-the-art providing a complete architecture that better considers the functional needs of mHealth apps.

4.3. PROCEDURE FOR ARCHITECTURE INSTANTIATION INTO SERVICES

The architecture outlined in the preceding section serves as a general blueprint to streamline the development of mHealth systems. When faced with a specific mHealth system use case, designers can instantiate the architecture by selecting the modules pertinent to their needs. This process involves four key steps, summarized in Fig. 4.

Firstly, designers embark on the initial step of describing the goal of their mHealth system (i.e., which service logic they want to implement) and identifying the target users, data sources, devices (both for sensing and interface), and specific privacy/security constraints. Subsequently, in the second step, designers delineate the functional requirements for

Table 3

Services of layers responsible for data management and protection. Notes on implementation include some potential technologies/protocols that can be used when instantiating the architecture for a specific use case.

Layer	Service	Fusion Features	Description	Notes on implementation
Patient Features Extraction	Multimedia processing service	FE1	Processing of multimedia data sent by the patient (e.g., pictures, videos, audios) or the clinician (e.g., X-rays) to accelerate the data analysis.	Includes DL methods, e.g. CNN for object detection and classification from images or video, and NLP for disease detection from audios.
	Biomedical data processing service	FE1	Processing of bio-signals (collected from IoT devices) or relevant data for the clinical use case (self-reported by the patient), to obtain interpretable features, or to be used in further services.	Includes ML/DL methods for variations detection in vital signals, e.g. body temperature or ECG.
	System usage processing service	FE1	Processing of data related to system usage, including actively submitted data and metrics about patient interaction with the app.	Hybrid learning combining statistical metrics and Markov Models (MM) can be used to model standard behavior and analyze outliers. Also, ML can be used to classify activity (e.g., Naïve Bayes).
	Context processing service	FE1	Processing of embedded sensor data to extract information about the context of app usage (e.g., location) or life patterns (e.g., steps, sleep patterns).	ML methods can be used to analyze phone sensor value (e.g., Decision Trees).
Data Intelligence	Inference engine service	DI2, DI3	It obtains new information based on the knowledge or combining different types of data.	AI algorithms (e.g., Regression algorithms, Decision Trees, Causal Graphical Models) or counterfactual analysis can be used to explain events, extracting causal relationships from data.
	Anomaly detection service	DI1, DI3	Detection of any data out of normal standards or with negative trend.	DL methods may be used, such as CNN, Isolation Forests, NLP, etc.
	Behavioral analysis service	DI1-DI3	Analyzes system usage or contextual data (e.g., location data) to obtain information about patient behavior.	Clustering techniques (e.g., DBSCAN, Naïve Bayes or Hidden MM) and predictive models (e.g., Logistic Regression) may be used.
	Health profiling service	DI3	Patient profiling in groups to obtain useful information about their state of health.	Clustering techniques to profile patients into severity groups or patterns of illness evolution.
Situation Assessment	Health state evaluation service	HA1	Health evaluation based on statistical analysis and classification algorithms.	ML algorithms can determine the health state using historical data (e.g., SVM).
	Risk analysis service	HA2	Detection of worsening or risk evolution.	Both statistical techniques and ML (e.g., SVM) CNN, NLP, etc.
Health Logic	Intelligence orchestration and rule engine	HL1	On the info from previous services (health state, risk level, adherence ...), it decides which actions to take. It coordinates analysis modules. It dispatches instructions to customize info display, data collection, reminders ...	Orchestration can be performed via REST or following event-based architectures. As for the rule-based execution, RulesEngine by Microsoft may be applicable. Alternatively, cloud-based or self-hosted lambda-like functions can be used to implement this service.
User-in-the-Loop Layer	Recommendation service	HL1	Delivery of recommendations to patients or clinicians on the health profile.	AI can be used for personalization, e.g. classification algorithms.
	Data visualization service	UL3	Dashboard components for information, monitoring and assessment.	AI tools for view customization. Use of multiplatform frameworks (e.g., Kibana or Grafana).
	P2P communication service	UL2	Communication between clinicians and patients (or caregivers).	WebRTC may be used to provide real time text, audio and video communications.
	Notifications & alerts service	UL1	Reminders and alerts delivery to the stakeholders (reminders of data input, alerts on risks, new message notifications, etc.).	Supabase or Firebase services can be used to generate PUSH alerts. Alternatively, SMS gateways services allow to distribute SMS messages.
Patient Adherence& Adaptive Acquisition	Adaptative data acquisition system	AD1, AC1-AC3	Adaptive sampling. Analysis of the quality and amount of data from sensors. Sensor collection status check-up.	Diagnostic tools for bad connection or insufficient permissions given to the app check-up.
	Adherence support service	AD1, AC1	Analysis of system usage to determine how to improve adherence and coordinate the needed actions (e.g., visualization changes, requests for information, etc.)	Clustering techniques can be used to identify adherence categories of users and customize actions to improve it.
User Management & Privacy	Authentication & authorization service	S1	Delivery of access to data when the user has permission, through the exchange of authentication keys and access credentials.	Custom-built, on standard protocol such as JWT or OAUTH or integrated with some external Identity and access management (IAM) services (Google Cloud IAM, Azure Active Directory, Okta).
	Users service	S1	User roles management and access to the user DB.	User Management Systems (UMS), often include the User DB, such as Active Directory and OpenLDAP. They are also provided by the several cloud-service providers.
	Users DB	S1	Database containing user personal data (e.g., name, age, telephone number ...).	In the custom-built solutions, they are generally implemented using a generic database.

each stakeholder, mapping and analyzing them with respect to the common *mHealth Fusion Features* (as outlined in Table 1). In the next step, designers proceed to select the needed services by identifying them in Tables 2 and 3 (third column) so that the identified features are fulfilled. Consequently, designers can make informed decisions by selecting the necessary services to that their mHealth architecture is instantiated as in Fig. 2. Then, Tables 2 and 3 also suggest potential implementations and AI algorithms that can be used in each service, guiding the final development and implementation of the mHealth system.

5. Case-based analysis: application of the architecture to practical scenarios

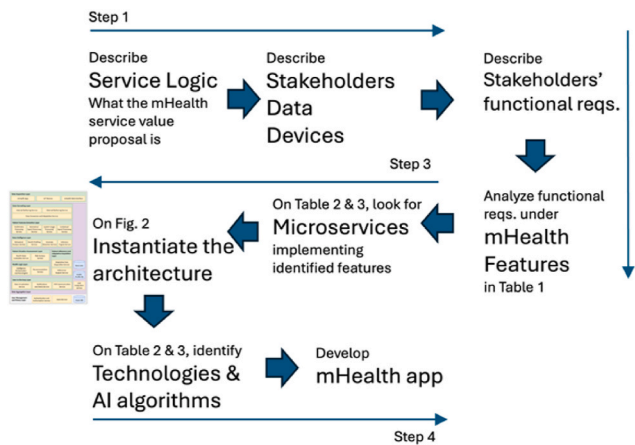
A use-case based validation has been carried out to probe the value and feasibility of the proposed architecture. The objective of this validation is three-fold.

- To demonstrate that architecture is feasible from a technical point of view. By instantiating and implementing the architecture in real use

Table 4

Comparative analysis of the proposed architecture with existing architectures in the literature.

Ref.	Architecture Objective	Similarities	Differences
[36]	Generic technological stack to develop m-Health services assisted with AI.	It includes data acquisition, aggregation, intelligence, and visualization. It does consider the usage of AI to analyze mHealth data.	Architecture components are technological services. It is not a functional architecture with health-related blocks. A comprehensive health analysis is not considered. Absence of clinician-patient communications.
[33]	Architecture to enable Body Area Networks using the mobile phone as a sensor hub.	Mobile phone is used as a gateway to aggregate health information from IoT sensors. A centralized/cloud platform is used for processing/visualization	Data intelligence is only considered generically without defining functional services for feature extraction, data intelligence, patient assessment, etc. Interaction between users is not considered.
[34]	Architecture with a focus on multi-tenant health data storage over health ontologies.	Elements to support decision-making/provide recommendations are considered. Data aggregation is considered using structured data bases.	Multi-tenant data storage is considered to improve security and privacy. Data acquisition is not considered. Functional services for data analysis are not specified.
[35]	Comprehensive mHealth architecture.	It covers multi-device data acquisition, data aggregation, processing, and health recommendations.	Privacy/security issues are not covered in this architecture. Functional data analysis blocks are not clearly identified. AI processing not considered.
[9]	Use-case oriented mHealth architecture for pregnancy follow-up.	It considers data acquisition from multiple devices, data aggregation in a common data format (ontologies) with integration with EHR, multi-device and multi-user data visualization.	Data processing is not covered within the architecture. AI processing not included.
[8]	Use-case oriented mHealth architecture for hypertension management	It considers data acquisition from IoT devices, user interaction including communication and data privacy features.	Data processing is not covered within the architecture. AI processing not included.

**Fig. 4.** Procedure for architecture instantiation to a use case.

cases, it is possible to prove that it is not a mere conceptual construct but a practical technical design that developed successfully with existing technologies.

- b) To demonstrate that the architecture fulfills common features of mHealth systems. By selecting an assorted set of use-cases covering a wide range of requirements, it is possible to prove that the architecture can be catered to fulfill the requirements of mHealth systems.
- c) To demonstrate that the architecture goes beyond the current state of the art architectures. Selected use-cases also serve as a comparison mechanism by analyzing whether they could be implemented with other existing architectures.

To support these objectives, the validation methodology described below has been applied. First, a selection of use cases has been performed. Specifically, four use cases were developed: *UC1) youth with mental health issues monitoring*; *UC2) SARS-CoV-2 patients monitoring*; *UC3) fluid intake control in kidney disease patients*; *UC4) peritoneal dialysis treatment monitoring*. The use cases are derived from collaborative efforts

Table 5

Step 1 – Initiation step for each case of use.

Use case	Users	Devices in use/UI	Health Logic
UC1-Mental health	Patient, therapist, family	Smartphone (location, apps usage, emotions). Web UI.	Improve therapist's follow-up of young patients under mental health treatment, and prevent risk, by collecting patient's daily emotions and lifestyle Enhance data security with patient's explicit authorization.
UC2-SARS-CoV-2	Patient, clinician	Smartphone (location, apps usage) Web UI	Monitor health status of large volumes SARS-CoV-2 patients, to improve the evaluation speed and prevent risks.
UC3 – Kidney disease	Patient, clinician	Smartphone (camera) Weight scale Web UI.	Fluid intake control of kidney disease patients to prevent failure or overhydration, by collecting drinks, meals and weight.
UC4-Peritoneal dialysis	Patient, clinician	Smartphone, smart band, weight scale, Web UI.	Improve dialysis patient follow-up to ensure the correct treatment adherence and to assist the decision making.

Table 6

Step 2 - Required features in each use case.

Use Case	Target users			Supp. devices		Security	Data acquisition		
	U1	U2	U3	D1	D2	S1	AC1	AC2	AC3
	Patient	Clinicians	Caregiver	Smartphone	Web access	Security & Privacy	Self-reported measures	Embedded sensors	IoT wireless devices
UC1	✓	✓	✓	✓	✓	✓	✓	✓	
UC2	✓	✓		✓	✓	✓	✓		
UC3	✓	✓		✓	✓	✓	✓	✓	✓
UC4	✓	✓		✓	✓	✓	✓		✓

between the authors and Spanish healthcare providers, as well as from the authors' involvement in mHealth EU innovation projects. They have been chosen because they address real needs and challenges identified by clinicians and patients/users, while presenting a diverse range of functionalities and requirements (useful for architecture validation). Second, the proposed mHealth architecture is instantiated for each use case by using the instantiation procedure described in Section 4.3: mHealth system description (step 1), mHealth features identification (step 2), architecture instantiation (step 3), identification of relevant AI algorithms and technologies for the implementation (step 4). Third, each use case is implemented in a real mHealth system and tested with real users. In this sense, it is not the aim of this article to report a clinical study on the impact of mHealth services on the users, or to analyze the performance of the implemented AI algorithms. The use cases and their implementations should be understood as evidence to probe that proposed architecture can serve, in practice, as a conceptual tool and guide the design of different mHealth solutions. Finally, a conceptual comparative analysis has been performed to analyze the feasibility of implementing each use case under other existing architectures.

5.1. Use case identification and description of the associated mHealth systems

As the use cases have already been identified, and following the design procedure in Section 4.3, the first step is to describe the service logic for the system, the target users and the devices to collect data from, together with specific security and privacy issues that may condition the structure of databases and components (Table 5).

Starting with UC1, its motivation stems from the observation that standard interventions for youngsters with mental health issues typically involve weekly meetings. However, therapists often find that the information gathered in a session is incomplete, making it challenging to detect high-risk situations effectively. To address this limitation and to better evaluate patient progress, an application was deemed necessary. This application would collect mood self-reports multiple times a day, alongside behavioral information derived from smartphone sensors and application usage. With each mood self-report, additional contextual data (about reporting context, sleep quality or digital activity) would be provided, allowing therapists to gain deeper insights into the patterns underlying certain feelings and enriching therapy. Furthermore, as an exploratory feature, a system triggering alerts on potential deviations over standard behaviors was also identified as relevant. A patient-enabled control system was required to enable data access exclusively during therapy sessions, ensuring that patients retain control over their information.

Regarding UC2, it was designed at the beginning of the SARS-CoV-2 pandemic when the healthcare system suffered an unmanageable care demand. The high risk of contagion made it difficult to monitor patients while reducing the workload of overburdened healthcare workers. A mHealth system would enable patients to report their symptoms and the clinicians to organize patients (maximizing the efficiency of monitoring) and easily track the progress of their condition, also including an alert system if their health condition deteriorates.

UC3 focuses on kidney diseases patients, who require a precise

control of liquid intake, as hyper-hydration can worsen the condition, being sometimes relevant to ongoing treatments. The proposed mHealth system would need to accurately measure the amount of fluid ingested per day and alert in case they should reduce it. Also, weight variation would be reported using an IoT smart weight to better estimate fluid retention.

Finally, patients under automated peritoneal dialysis (UC4) use a machine (a cycle) at home to fill and drain the abdomen, being required to record all their fluid drainage and infusion activities throughout the day. The proposed mHealth service is oriented to support the record of treatment data and relevant health parameters retrieved from IoT devices (a smartwatch and a smart scale - e.g., weight or blood pressure), enabling evolution follow-up by clinicians. Additionally, the system could facilitate information, communication, and management of unexpected evolution. Moreover, improving adherence would be desired, so to improve patients' commitment with their own health and their treatment.

5.2. Architecture instantiation per use case

With the background information, it is possible to proceed with the second step of the instantiation procedure: identifying the general mHealth features deemed as necessary for each use case. The result of this process is depicted in Table 6, which shows that while each use case individually covers different mHealth needs, the selection of use cases ensures that all features are at least covered once. Then, step 3 of the procedure is performed, selecting which microservices are needed in each use case, resulting in a different architecture per use case. For brevity, these two steps are discussed jointly hereafter.

Taking Table 2 as a reference for the functionalities identification, several features are common to all cases such as patients (U1) and clinicians (U2) and the use of smartphone (D1) and web access (D2). Only in the case of mental health caregivers (family members) are included (U3). Moreover, all cases include patient reported information (AC1), but only some of them require retrieving information from smartphone sensors (gyroscope, accelerometer, light detection, location) or virtual sensors (use of mobile applications, mHealth service usage) or IoT devices (AC2, AC3). To achieve this, it is necessary include the *mHealth app* and *mHealth web interface* to the system using *internal collection service* and *data conversion and adaptive service* from the Data Harvesting Layer. The integration of IoT devices, in UC3 and UC4, is achieved through Bluetooth connectivity with the app, so none of them implements the *external collection service*. The EHR integration service is included in UC3, as it requires to be integrated into the clinical process to provide data to a hospital through the HL7 FHIR standard.

Data pre-processing (FE1) is always required, but its implementation depends on the scenario. The *biomedical data processing service* is present in all our systems to handle mood states, symptoms reported by the patient, intakes reports, treatment information, IoT data, etc. *Contextual data processing service* is used in UC1 and UC3 systems to process embedded sensor data, such as sleep interruptions detected by the mobile phone or physical activity. The *multimedia processing service* is used for the image collected in liquid intake system and *system usage processing service* processed the app interaction data in UC4.

Data aggregation		Feature extract.	Data intelligence			Health sit. asses.	Health logic	Adhe-rence	User-in-the-loop		
AG1	AG2	FE1	DI1	DI2	DI3	HA1	HL1	AD1	UL1	UL2	UL3
Data integration	HER EHR interoperability	Data pre-processing	Anomaly detection	Inference engines	AI solutions	Health state evaluation	Rule-based functionality	Adherence support	Notifications	Communication	Data visualization
✓		✓	✓	✓	✓		✓	✓	✓		✓
✓		✓	✓		✓	✓	✓	✓	✓	✓	✓
✓	✓	✓	✓	✓	✓		✓	✓	✓	✓	✓
✓		✓	✓	✓	✓	✓	✓	✓	✓	✓	✓

The extracted features are used in different services of the following layers. In some cases, features are extended with an *inference engine* (DI2). For example, in UC1, by combining mood reports, screen lock state, accelerometer and light detection data, the system is able to extract approximate mobile phone interaction patterns and patient sleep interruptions. In UC3, the data is combined to obtain the total liquid intake. In addition, the *anomaly detection service* (DI1) is used in all the systems, which also helps the implementation of the *risk detection service* to support patient follow-up and prevent worsening. The health profiling service can be included to classify patients into groups relevant to the case: severity of SARS-CoV-2 (UC2) or level of adherence to dialysis app (UC4). These profiling groups are based on a system of rules (e.g., fever threshold) and classification algorithms (e.g., k-Means, Logistic Regression and SVM). The obtained information serves the Health Logic Layer. Rule engine service (HL1) is also included in the four use cases, for notifications firing or visualization adaptation. This latter adaptation is used to support the assessment of the patient state and the monitoring efficiency (e.g., selecting charts/indicators to quickly interpret the evolution or displaying first evolution of critical patients). In some cases (UC1 and UC2), the recommendation service displays messages to the patient and/or the clinician depending on patient state.

Only UC4 includes the Patient Adherence and Adaptive Acquisition Layer as it involves patients with a long-term disease, and the main objective is to improve users' adherence to their treatment. To achieve this, the adherence support service follows user profiles to act in different ways, such as by increasing the number of notifications reminding the patient to enter data. Also, adaptive data acquisition service is used to adjust the frequency of data collection by the smart wristband.

Regarding the User-in-the-Loop Layer, all these scenarios require the data visualization service (UL3) to display at least the patient's status to the clinician, and the notification and alert service (UL1) to remind the patient to enter data and alert the clinician of any detected problem. Additionally, some use cases (UC2, UC3 and UC4) implement a p2p communication service (UL2), which is especially critical for cases of home confinement due SARS-CoV-2 (UC2).

Finally, it is mandatory to include security and privacy (S1), which is developed in the authentication and authorization service. In UC1, the access to patient's data could be enabled in real time but due to the sensitivity of the data privacy, the service requires explicit and real-time authorization from the patient (via a temporary access code). This authorization is granted to the therapist for an ephemeral period (for example, during face-to-face sessions). Meanwhile, UC2 focuses on assigned roles to registered users, within a hierarchy (patient, clinician type and supervisor), simplifying patient management by ensuring users access only to needed data (clinicians only access their patients and a supervisor manages the patients assigned to each clinician).

As a summary of the instantiation process Fig. 5 illustrates the blocks used in each use cases architecture (identified by colored and numbered circles for each use case). It also proves how the proposed mHealth architecture can be easily catered to fulfill a wide variety of common mHealth features.

5.3. Implementation of mHealth systems and real user tests

Moving to the final step of the procedure, different technologies and protocols have been used in the implementation of each mHealth system considering the recommendations provided in Tables 2 and 3 RESTful communication has been applied to the frontend and mobile applications, as well as for the majority of backend module communication. For database communication, TCP/IP is used in conjunction with the specific protocol of each database type (MySQL, MongoDB and Redis). For asynchronous communication, the choice is Firebase (over APN for iOS) for sending notifications to mobile applications and WebSockets for browser-based frontend notifications. Finally, MQTT or Kafka are employed for asynchronous backend communication (e.g., to notify the

result of AI inference to consumer services). Similarly, different ML and DL algorithms are specifically considered in each system. These algorithms are identified in Table 7.

The resulting mHealth systems have been deployed and tested with real users. In UC1, microservices and databases (backend) were deployed entirely in the cloud, using scalable and secure cloud-based resources (AWS). UC1 included the monitoring of 46 volunteers during the period between October 2020 and March 2021 [80]. Regarding UC2, the backend components were deployed in the cloud using a public cloud service. The system was deployed for patient follow-up during the spring of 2020, with a total of 38 enrolled patients [81]. For UC3, the final deployment was a hybrid solution of federated clouds and on-premises servers: each partner of the project deployed its software components in a different location and all of them were able to read and write specific data from the others. It is noteworthy that, in this use case, the architecture has also been integrated with an external hospital information system based on EHR, using the FHIR standard for the communication and the data model. As part of a project [82], the app was released on Google Play for healthy individuals (not as a clinical product), but there was no subsequent follow-up of the users who installed and used it. Finally, the deployment of UC4 case is completely on-premises, on servers owned and maintained by our research group. This system is currently being used by 13 volunteers and being prepared to carry out a broader clinical study [83]. Fig. 6 shows some examples of the resulting products. Overall, the implementation and deployment of each use case shows the technical feasibility of the proposed architecture.

5.4. A comparative analysis with existing architectures

To conclude the validation methodology outlined above, a conceptual analysis is carried out to determine whether the proposed use cases can be implemented with existing architectures. In this sense, a general comparison has already been provided in Section 4.2. A first barrier for the implementation of the use cases with the analyzed architectures ([8, 9,33–36]) would be the lack of mHealth specific services and AI processing. For instance, as already discussed, UC1 requires a service to analyze patient behavior as a precondition for the health logic of the use case. Similarly, user profiling is to be used in UC2 or health risk analysis is required in all UCs. In this sense, the lack of clearly defined modules may not constitute a hard limit for the adoption of literature architectures, but it hinders the development process. A major handicap would be the lack of user interaction services for UC2-4 or the absence of adherence features in the considered architectures.

Overall, the inexistence of a comprehensive architecture in the current state of the art means that the full implementation of the analyzed use cases would be challenging. In this context, the architecture proposed in this paper represents an advance over the current state of the art, enabling the easy development of new use cases in mHealth services.

6. Conclusions

This article delves into the development of versatile mHealth systems, aiming to address the lack of standardized structure and components in their design. It presents a comprehensive architecture rooted in data fusion and AI, consisting of eight modular layers with customizable microservices tailored to specific system requirements. Additionally, two complementary layers are incorporated to ensure efficient data aggregation and privacy protection. The adaptable AI modules leverage data to enrich the capabilities of both patients and healthcare professionals.

The architecture is the result of combining a bottom-up functional analysis approach with insights from existing literature on mHealth systems, complemented by a customized adaptation of the JDL model. This approach is thought to lead to cost-effective developments, solutions with better functional extensibility, simplified system

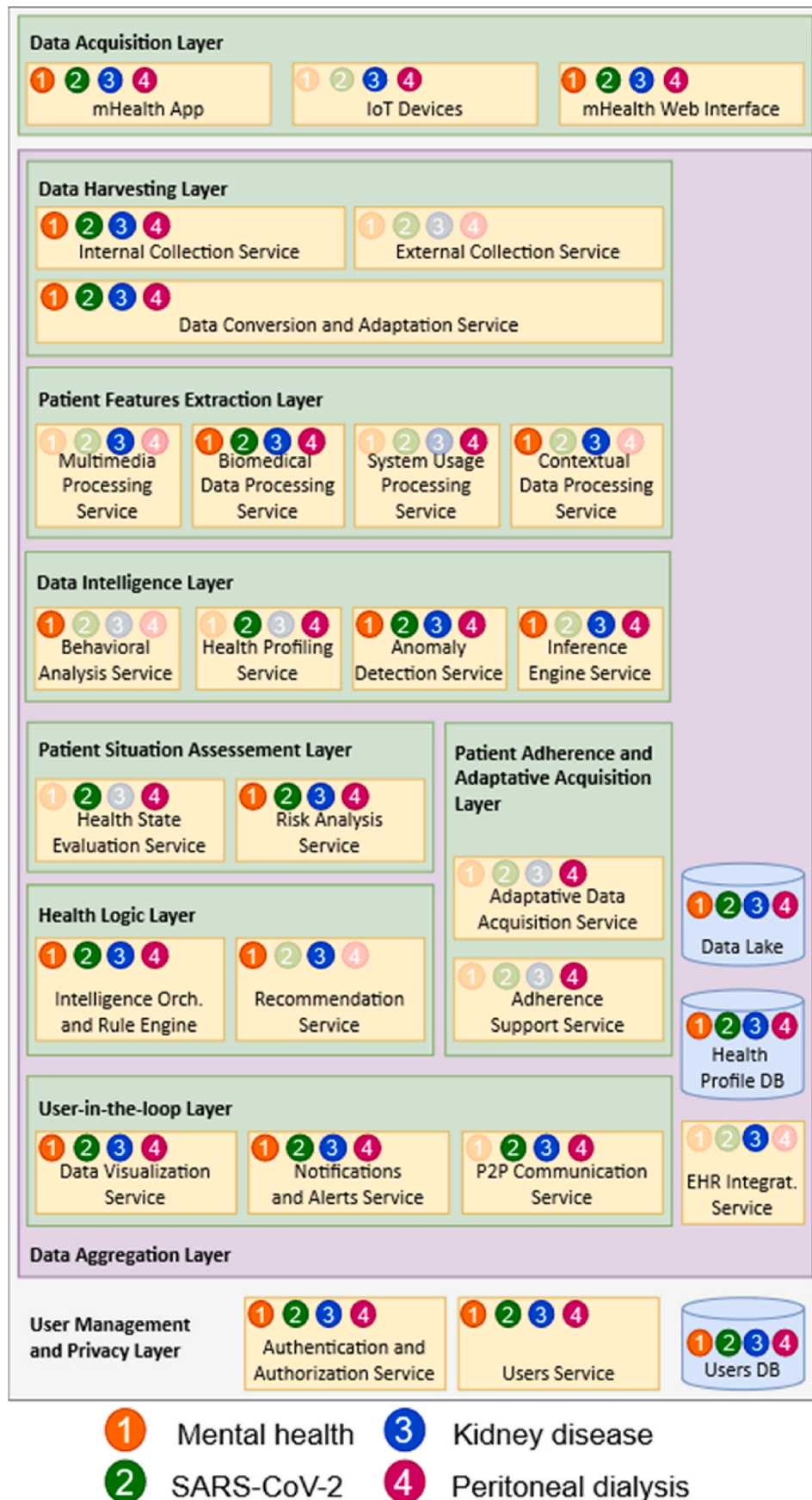


Fig. 5. Architecture instantiations of each use case.

Table 7
AI algorithms used in each use case.

Use case	AI algorithms
UC1	- Decision tree-based classifier to related mood with physical, sleep and digital behaviour. - Spatial clustering pipeline to build a baseline behavior model based on frequent stay points. - Naïve Bayes predictor to label frequent stay points as non-frequent and analyze anomalies. - XGBoost classifier to label daily digital behavior as anomalous.
UC2	- Rule engine based on indicators to detect anomalies in patient trends.
UC3	- Image analysis with CNN to identify food type and water volume, to determine water consumption from meals photographs.
UC4	- k-Means clustering on app usage (time in app, notifications read, number of user inputs, number of messages to clinicians, etc.) to create profile-based adherence strategies.

maintenance, and improved knowledge management alongside personalized systems. Its validation has been conducted through four distinct use cases related to health logic, providing evidence of its effectiveness. While current systems have been predominantly vertical and focused on specific conditions, this architecture facilitates the development of horizontal systems capable of extending the use of the system to tackle evolving comorbidities in a cohesive manner. However, validation of these more advanced and less common use cases remains an ongoing endeavor.

While security mechanisms are conceptually integrated into the proposed architecture, protection against cyberattacks should be considered. For instance, safeguarding from data poisoning (when data is deliberately manipulated to train ML models) could be achieved through solutions such as blockchain, in which a decentralized network of nodes (e.g., in patient smartphone) would validate and confirm

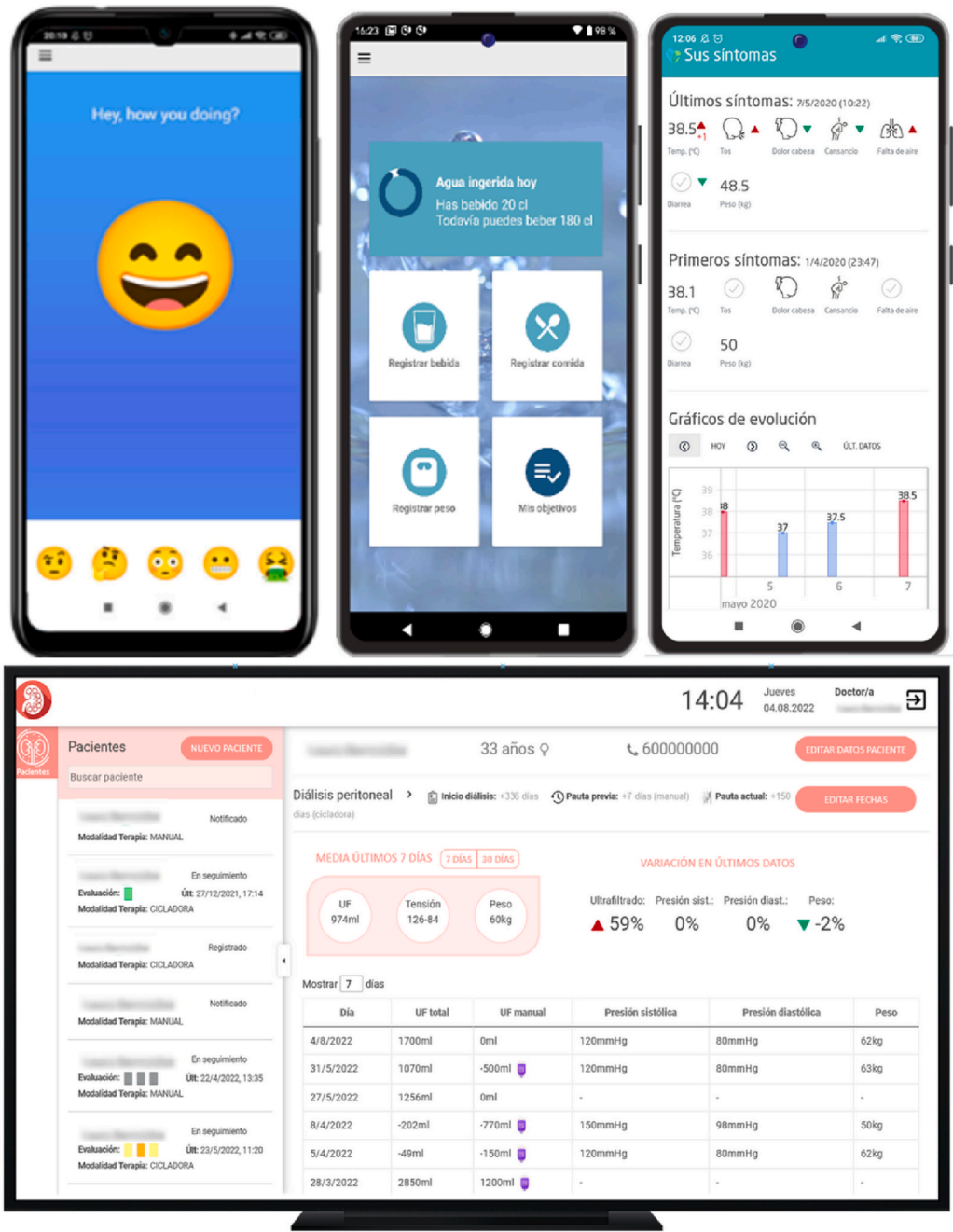


Fig. 6. Architecture instantiations for each use case and AI algorithms in use. From left to right: reporting patient mood (UC1), home menu with patient options and recommendations (UC2), patient’s historical symptoms (UC3) and professional web interface for managing patients (UC4).

transactions through a consensus mechanism. Furthermore, in terms of regulation, different regions or countries may have varying standards governing mHealth systems and services, leading to compliance challenges that are not directly addressed in the architecture.

The validation scenarios considered in this article are small scale ones. Scalability testing of the proposal with a large user base was not the objective of this research but may pose some specific challenges related to data storage or model sharing. Another issue is the integration of data from external providers, such as vendor-locked devices, usually solved with the collaboration of the patient (to input data directly or to take pictures of the screen for posterior analysis, etc.). Additionally, the scenarios analyzed in Section 5 did not require full connectivity to clinical systems. In practice, the integration of mHealth with EHR is usually bypassed to accelerate system development, making clinicians to work with different systems and duplicating their work (e.g., data input in each system). How to deal with associated data to EHR that can evolve out of the system, but in a coordinated manner, is still an open issue.

As a further step, we consider making an exhaustive analysis of which AI elements could be readily incorporated for mHealth system implementation in an off-the-shelf manner. This may encompass techniques for advanced feature extraction, mechanisms to enhance adherence, detect anomalies or establish baseline patient behavior on certain indicators that may run in a distributed or centralized manner. Additionally, we find that personalization of mHealth can improve adherence to the system and improved benefits, so a review of the architecture to deliver individual mHealth systems is also in our agenda. The use of federated learning to train models in patients' devices or to aggregate models at different levels (groups of patients, hospital, region, etc.) may be a solution to support personalization, while mitigating the risk for cyberattacks, preserving privacy and knowledge sharing.

It is crucial to recognize that the adoption of mHealth has profound implications for clinical care and requires careful consideration of its structural and organizational aspects. A thorough analysis of the clinical pathway is imperative when implementing an mHealth solution, involving the identification of stakeholders and required competencies. Our future roadmap also includes the development of a service design methodology to better understand the deployment implications of mHealth services, along with formal validation of mHealth impacts across the entire value chain.

CRedit authorship contribution statement

Ana González Bermúdez: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Investigation, Formal analysis. **David Carramiñana:** Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Investigation, Formal analysis. **Ana M. Bernardos:** Writing – review & editing, Writing – original draft, Validation, Supervision, Project administration, Methodology, Funding acquisition, Conceptualization. **Luca Bergesio:** Writing – review & editing, Writing – original draft, Validation, Software, Resources. **Juan A. Besada:** Writing – review & editing, Writing – original draft, Supervision, Methodology, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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